

Domain / Parameter	Symbol / Variable	Analytical / Simulated Value	Units	Design Significance & Boundary Context
Kinematics (Relative to Stance Ankle)				
Anterior Payload Offset	XP/foot	+0.1286	m	Places payload CoM anterior to stance ankle, counteracting uphill pitch
Vertical Elevation of CoM	YP/foot	+1.3073	m	Maintains high thoracic center of mass, reducing cervical strain
Kinematics (Relative to Rear Wheel)				
Horizontal Trailing Baseline	XL	-0.6086	m	Extended mechanical footprint behind wearer for pitch stability
Vertical Height Above Contact	YL	+0.7911	m	Strut height clearance for uneven rocky obstacles
Resultant Payload Vector	// PL //	0.9981	m	Closed-loop 5-link chain terminal vector magnitude
Suspension & Dynamic Interface				
Terrain-Normal Component of Payload	Floadcosθ	368.74	N	Available normal gravitational load budget along 20° incline
Active Hinge Internal Resistance (Static)	Fdamp,static	106.6	N	Preload force supported across harness interface at rest
Active Hinge Internal Resistance (Peak)	Fdamp,dyn	148.5	N	Peak damper-spring reaction at heel-strike (x'=0.60 m/s)
Transmitted Ground Normal Force (Static)	Fground,static	269.28	N	Static normal load offloaded into the terrain through omni-wheel
Transmitted Ground Normal Force (Peak)	Fground,dyn	418.55	N	Peak impact transient supported by terrain via suspension
Static Ground Shunt Efficiency	ηshunt,static	71.68	%	Ratio of static terrain-normal payload diverted from human
Dynamic Peak Shunt Efficiency	ηshunt,dyn	73.8	%	Shock isolation efficiency during dynamic impact phase
Traction & Ground Interaction				
Tangential Downslope Ground Reaction	F //	97.99	N	Longitudinal gravity component resisted at contact patch
Required Friction Coefficient	μreq	0.364	—	Derived from tan(20°); minimum coefficient to avoid slip
Terrain Safety Factor Against Slip	SFtraction	1.65	—	Evaluated against dry rock (μ=0.60); confirms non-slip rolling
Biomechanical Skeletal Relief				
Nominal External Payload Weight	Wp	400	N	Total carried cargo weight (40.77 kg×9.81 m/s²)
Total Combined System Weight	Wtotal	1135.75	N	Sum of porter (75.0 kg) and payload gravitational loads
L5/S1 Compression (Unassisted)	FL5/S1,unassisted	828.5	N	Upper body weight (428.5 N) + full payload (400 N)
L5/S1 Compression (LoCaDe-Exo)	FL5/S1,assisted	548.5	N	Upper body weight (428.5 N) + residual payload (120 N)
Net Spinal Load Reduction at L5/S1	ΔFL5/S1	33.8	%	Total intervertebral compressive relief on the lumbar spine